PEDIATRIC CARE

KPPC Multisystem Inflammatory Syndrome in Children (MIS-C) Pathway for Stable, Non-Toxic Patients[1]

If the child (age 0-18) stable and non-toxic[2] and has:

Fever 38C or higher for 1-2 days (can be subjective) AND has 2 or more of the clinical features AND Has a known close COVID contact[3], primary illness or COVID vaccine in the last 12 weeks.

MIS-C Definition

PTAC
562-658-3980
8-137-3980

Clinical Features for MIS-C[4]

1. GI symptoms-diarrhea, emesis, abdominal pain.

2. Mucocutaneous-rash, non-exudative conjunctivitis, erythematous cracked lips, strawberry tongue, erythema of oral or pharyngeal mucosa, Erythema or edema of hands or feet.

3. Shock/hypotension

4. Neurologic-headaches, irritability, lethargy, altered mental status, paresthesia, lethargy.

Differential

YES

NO

Patient should be followed up in clinic in 24-48 hours if still febrile, clinically worsening, or developing further symptoms

Obtain 1st Tier labs:
1. CBC with Diff
2. Lytes, BUN, creatinine
3. CRP and ESR
4. COVID PCR swab
5. COVID serology
6. BCX, UA, UCX

Abnormal Labs:
CRP >30 mg/L
ESR > 40 mm/hr
Na < 135 mEq/L
Plt < 150,000/mcl
ALC < 1000/mcl
ANC > 8000/mcl

NO

YES

- If the suspicion for early MIS-C necessitates further guidance for next steps, including 2nd tier testing, hospital observation, or follow up plans, contact local Pediatric ID specialist or use PTAC to contact Pediatric Hospitalist for recommendations.

- If 2nd Tier testing is recommended, would consider:

1. BCX, UA, UCX if not already collected
2. EBV serology, ASO, Bacterial GI panel if predominantly GI symptoms
3. Albumin, CK, AST, ALT, T/D bili, PTT, INR
4. Triglycerides, LDH
5. Ferritin
6. BNP and troponin (new high sensitivity Troponin is abnormal if >30 pg/mL)

Outpatient-contact PTAC to arrange transfer if admission indicated
Unstable, Toxic Patient-go to Inpatient, ICU and ED pathway

1Patients should be seen in person in clinic for evaluation

2For sick/ill appearing patient, please refer to "ill" pathway

3Close COVID contact = <6ft for >15 minutes

4Please consider differential diagnosis. Note that classic and Incomplete Kawasaki Disease (KD) should strongly be considered in children presenting with 5 or more days of fever and 4-5 of the clinical criteria for classic KD, or with 2-3 clinical criteria for incomplete KD, as well as infants with 7 or more days of fever with no other explanation. *Denotes the Kawasaki Disease clinical criteria above.

Stable Pathway approved 6.30.2020, Updated 8.3.2020, 12.14.2020, 11.16.2021
Subcommittee: Dr. Paul Han, Dr. Mark Salzman, Dr. Zackary Taylor, Dr. Evan Steinberg, Dr. Judith Garza, Dr. Anusha Ramanathan, Dr. Rhina Castillo, and Dr. Joseph Mares

Unstable Pathway

PEDIATRIC CARE

INPATIENT, ICU AND EMERGENCY DEPARTMENT PATHWAY FOR MULTISYSTEM INFLAMMATORY SYNDROME (MIS-C)-UNSTABLE PATIENTS

If the child (age 0-18) ill-appearing and has:

Fever 38C or higher for 1-2 days (can be subjective) AND has 2 or more of the clinical features AND Has a known close COVID contact[3], primary illness or COVID vaccine in the last 12 weeks.

Consider differential

Treat possible sepsis/ shock
-ABCDEs, Vitals, O2
-IV access-2 large proximal PIV, IO if unable to obtain PIV
-Obtain Tier 1 and 2 labs and those based on differential
-Consider POC echo if able

Fluids (if unstable):
1. Bolus 20 ml/kg crystalloid – UNLESS concerned for cardiogenic shock (poor function on bedside focused echo, crackles, hepatomegaly, cardiomegaly on CXR)
2. If concerned for cardiogenic shock give 5-10 ml/kg bolus and IMMEDIATELY reassess (w/in 15 min) and repeat if no sign of worsening perfusion
3. Consult intensivist if not improving to initiate inotropes
4. Consider need for ECMO in refractory shock or unstable arrhythmias not responsive to standard medical therapy or pressors

Antibiotics (after obtaining blood culture): Ceftriaxone/Clindamycin +/- Vancomycin

Contact pediatric intensivist via PTAC for advice and to arrange transport to KPPC Tertiary Center

Consults as indicated:
Pediatric intensivist
Infectious Disease
Cardiology
Rheumatology

Clinical Features for MIS-C

1. GI symptoms-diarrhea, emesis, abdominal pain.

2. Mucocutaneous-rash, non-exudative conjunctivitis, erythematous cracked lips, strawberry tongue, erythema of oral or pharyngeal mucosa, erythema or edema of hands or feet.

3. Shock/hypotension

4. Neurologic-headaches, irritability, lethargy, altered mental status, paresthesia, lethargy.

Patient Transferred

Inpatient Management

Obtain Studies
1. Echocardiogram-within 24 hrs, or urgently if concerns of shock*
2. Chest X-ray
3. EKG*
4. Ensure all needed 1st and 2nd Tier labs obtained

Medications:
1. IVIG 2g/kg (to max of 100g)*
2. Solumedrol (discuss with Rheumatology on dosing)

Consults:
1. Rheumatology
2. Infectious Disease
3. Cardiology-if significant coronary artery involvement, LV dysfunction, arrhythmia, or if clinical myo/pericarditis
4. Neurology/Nephrology as indicated

Trend labs:
1. Repeat CBC, manual differential, CRP, lytes, Albumin, Ferritin, AST/ALT daily.
2. Trend Troponin/BNP daily if abnormal until improved OR can do q48h until afebrile, if initial set normal
3. Can space out labs to q48h if considerably downtrending and patient afebrile with clinical improvement
4. If requested, Rheumatology may also ask for D-Dimer and Fibrinogen

Anticoagulation:
1. Aspirin 3-5mg/kg daily to max of 81mg daily as long as platelets >80k
2. Lovenox *may* be recommended by hematology (eg if thrombus or high risk of thrombosis per hematology) or by cardiology.* If Lovenox recommended, start with goal anti-Xa level of 0.5-1 and continue as outpatient per heme/cardiology recommendations.

Discharge pending improvement in labs x48hrs, clinical improvement and afebrile x48hrs and tolerating oral steroids*

*See Inpatient Management for further details

Return to Stable Pathway

PTAC
562-658-3980
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APPENDIX 1: CASE DEFINITION OF MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN (MIS-C), LEVELS DIAGNOSTIC CERTAINTY*

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Level 1 of Diagnostic Certainty – Definitive Case
Age < 21 years (MIS-Ca)) OR ≥21 years (MIS-A)
AND
Fever ≥ 3 consecutive days
AND
2 or more of the following clinical features:
-Mucocutaneous (rash, erythema or cracking of the lips/mouth/pharynx, bilateral nonexudative conjunctivitis, erythema/edema of the hands and feet)
-Gastrointestinal (abdominal pain, vomiting, diarrhea)
-Shock/hypotension
-Neurologic (altered mental status, headache, weakness, paresthesias, lethargy)
AND
Laboratory evidence of inflammation including any of the following:
-Elevated CRP, ESR, ferritin, or procalcitonin[b]
AND
2 or more measures of disease activity:
-Elevated BNP or NT-proBNP or troponin[b]
-Neutrophilia, lymphopenia, or thrombocytopenia[b]
-Evidence of cardiac involvement by echocardiographyc or physical stigmata of heart failure[d]
-EKG changes consistent with myocarditis or myo-pericarditis[e]
AND
Laboratory confirmed SARS-CoV-2 infection[f]
OR
Personal history of confirmed COVID-19 within 12 weeks
OR
Close contact with known COVID-19 case within 12 weeks
OR
Following SARS-CoV-2 vaccination[g]

See footnotes on the following page

Level 2 of Diagnostic Certainty – Probable Case
Level 2a
Same criteria as Level 1 except:
1 measure of disease activity
AND
Within 12 weeks of a personal history of known or strongly suspected COVID-19
OR
Within 12 weeks of close contact with a person with known or strongly suspected COVID-19
OR
Following SARS-CoV-2 vaccination[g]
Level 2b
Same criteria as Level 1 except:
Fever lasting 1–2 days and can be subjective.
Level 3 of Diagnostic Certainty – Possible Case
Level 3a
Age < 21 years (MIS-C) OR ? 21 years (MIS-A)
AND
Fever ≥ 3 consecutive days
AND
2 or more of the following clinical features:
- Mucocutaneous (rash, erythema or cracking of the lips/mouth/pharynx, bilateral nonexudative conjunctivitis, erythema/edema of the hands and feet)
- Gastrointestinal (abdominal pain, vomiting, diarrhea)
- Shock/hypotension
- Neurologic (altered mental status, headache, weakness, paresthesias, lethargy)
- Physical stigmata of heart failure: gallop (IF diagnosed by expert) or rales, lower extremity edema, jugular venous distension, hepatosplenomegaly
AND
No laboratory markers of inflammation or measures of disease activity available
AND
Within 12 weeks of a personal history of known or strongly suspected COVID-19
OR
Within 12 weeks of close contact with a person with known or strongly suspected COVID-19
OR
Following SARS-CoV-2 vaccination[g]
Level 3b:
Same criteria as Level 2a except:
Fever lasting 1–2 days and can be subjective.
Level 4 of Diagnostic Certainty – Insufficient Evidence
Reported MIS-C/A with insufficient evidence to meet Level 1–3 in the case definition.
Example:
2 clinical features and history of COVID-19 within 12 weeks, but laboratory results and measures of disease activity are not available, and the fever criteria is not met.
Level 5 of Diagnostic Certainty – Not a case of MIS-C/A
Sufficient clinical and laboratory evidence exists to ascertain that a case is NOT MIS-C/A.
An alternative diagnosis has been ascertained

*Multisystem inflammatory syndrome in children and adults (MIS-C/A): Case definition & guidelines for data collection, analysis, and presentation of immunization safety data. Vogel et al. Vaccine 39 (2021) 3037-3049. Document available at: https://brightoncollaboration.us/wp-content/uploads/2021/07/MIS-CA-vaccine-publication.pdf

APPENDIX 1: CASE DEFINITION OF MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN (MIS-C), FOOTNOTES

Footnotes*:

Note: At all levels of certainty, minimal to mild respiratory symptoms may be present and their presence does not exclude a case of MIS-C/A, however, a case must be excluded if there is concern for acute COVID-19-related pulmonary disease. Further, one of the critical components of the case definition is that it is only applied when there is no clear alternative diagnosis for the reported event.

A MIS-C = multisystem inflammatory syndrome in children, MIS-A = multisystem inflammatory syndrome in adults, CRP = C reactive protein (detected by any measure),

ESR = erythrocyte sedimentation rate, BNP = brain natriuretic protein, NT-proBNP = N terminal pro-BNP, EKG = electrocardiogram, SARS-CoV-2 = severe acute respiratory syndrome coronavirus-2, COVID-19 = coronavirus disease 2019.

B Laboratory values are defined as low or high based on local laboratory normal ranges.

C Echocardiographic signs: dysfunction, wall motion abnormality, coronary abnormality (dilation, aneurysm, echobrightness, lack of distal tapering), valvular regurgitation, pericardial effusion, evidence of abnormal left ventricular strain.

D Physical stigmata of heart failure: gallop (IF diagnosed by expert) or rales, lower extremity edema, jugular venous distension, hepatosplenomegaly.

E EKG changes consistent with myocarditis or myo-pericarditis: abnormal ST segments and/or arrhythmia and/or pathologic Q waves and/or AV conduction delay and/or PR segment depression and/or low voltage QRS. f Laboratory evidence of SARS-CoV-2 infection: serologic evidence of SARS-CoV-2 infection or SARS-CoV-2 nucleic acid amplification positivity or SARS-CoV-2 antigen positivity.

F If a known or suspected COVID-19 infection has not occurred within the preceding 12 weeks.

*Multisystem inflammatory syndrome in children and adults (MIS-C/A): Case definition & guidelines for data collection, analysis, and presentation of immunization safety data. Vogel et al. Vaccine 39 (2021) 3037-3049. Document available at: https://brightoncollaboration.us/wp-content/uploads/2021/07/MIS-CA-vaccine-publication.pdf

PEDIATRIC CARE

APPENDIX 2: DIFFERENTIAL DIAGNOSIS*

DIFFERENTIAL DIAGNOSIS	CLINICAL FEATURES
Septic Shock	Can present with fevers, depending on infectious agent may have rashes, etc. This is more common than MIS-C, so should not be missed. It is important to suspect sepsis in a patient with tachycardia out of proportion to the degree of fever, and evidence of decreased peripheral perfusion. Hypotension is a late finding of septic shock in children.
	Patients appear clinically ill
	Associated with vital sign abnormalities, poor perfusion, and mental status changes
	Evaluate for focal source, e.g., skin/soft tissue, meningitis, pneumonia
	Obtain appropriate bacterial cultures
Toxic Shock Syndrome	Staph and strep toxin induced condition with
	Fever
	Vomiting/diarrhea
	Rash
	Confusion
	Conjuncitivits
	Headache
	Hypotension
Scarlet Fever	Fever
	Sore and red throat, some times with exudates, difficulty swallowing
	Enlarged glands
	Nausea, vomiting
	Headache
Steven Johnson Syndrome	Syndrome secondary to infection or reaction to some medications
	Fever and flu-like symptoms initially
	Widespread skin pain with red rash
	Blisters on skin and mucus membranes
	Red, watery eyes often with pain
Kawasaki's Disease	Commonly presents in children under the age of 6 and share many clinical features of MIS-C
	Fever with notable irritability
	Rash
	Non-purulent conjunctival injection
	Cervical lymphadenopathy
	Extremity changes including edema
Viral Infections	Nonspecific presentation of MIS-C may overlap with many viral illnesses
	Adenovirus
	Coxsackie/Enterovirus (particularly in the summer months)
	EBV
	HHV6 or HH7 (which cause Roseola) in infants/young children
Staph Scalded Skin Syndrome (SSSS)	Common in children under the age of 6
	Significant skin pain and erythema of the skin folds
	Generalized erythema of the skin as the disease progresses
	Flaccid bullae with erosions at the area of friction
	Fever, irritability and malaise
Myocarditis	Clinical signs of cardiogenic shock (cold extremities, diminished pulses/capillary refill, hypotension)
	Elevated troponin
	ST changes on EKG
	Cardiac dysfunction on echo
	Often associated with arrhythmias
Serum Sickness/Drug Reaction	Commonly presents following exposure to a drug or therapeutic agent
	Fever
	Rash
	Arthralgias
Fever and GI symptoms	Consider appendicitis, pneumonia, colitis, etc.

Return to Stable Pathway

Return to Unstable Pathway

*Modified from https://www.chop.edu/clinical-pathway/multisystem-inflammatory-syndrome-mis-c-differential-diagnosis

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APPENDIX 3: ED DISPOSITION, INITIAL MIS-C TREATMENT

Treatment if clinical syndrome and exam (+ labs if obtained) fit with MIS-C:

- Supportive care
- Begin with judicious fluid bolus (begin with 5-10ml/kg NS bolus) and assess frequently following each bolus
  - Unless the clinical suspicion is very high at the outset, many children are likely to get a 20 ml/kg bolus which is very appropriate for shock resuscitation.
  - Crackles; hepatomegaly; elevated BNP or Troponin (if available) or cardiomegaly on CXR should prompt providers to minimize fluids.
  - With septic shock fluid bolus improves tachycardia / hypotension and children should be fluid resuscitated aggressively and require frequent clinical reassessment when administering fluid boluses.
  - However, if improvement not seen (which occurs with MIS-C) consider cardiogenic shock. Inotropes should be discussed with cardiology / ICU sooner rather than later
- Multi-disciplinary team approach (PICU, pediatric rheumatology, cardiology, ID)
- Antibiotics after cultures pending confirmation of MIS-C
  - Ceftriaxone + Clindamycin +/- Vancomycin

Disposition in the ED:

- **If clinical suspicion high for MIS-C AND Ward admission planned:**
  - Early discussion recommended with pediatric hospitalist and subspecialists as necessary by contacting PTAC
  - Video teleconsultation if needed and transfer arrangements can be made by contacting PTAC. PTAC will connect referring center with pediatric center hospitalists and/or transport physician based at LAMC, DWN, ANA, and FON (SD area use current arrangements to reach pediatric hospitalists).
  - Determination of level of transport will be made at time of consultation with transport physician/hospitalist and referring physician. Unlikely to use BLS; ALS or higher. KP Transport Teams based at Fontana and Downey.
- **Pediatric ICU admission planned:**
  - Early discussion recommended with pediatric intenstivist and subspecialists as necessary by contacting PTAC
  - If troponins are elevated, then likely needs admission to PICU (high sensitivity Troponin of > 30 pg/mL)
  - Video teleconsultation with the tertiary center (LAMC, DWN, FON) can be obtained by contacting PTAC who will connect referring center with pediatric intensivist when assessing these patients and arranging for pediatric transport.
  - Discussion regarding mode of transportation will be agree upon (likely CCT-RN or KPPC Transport Team)
- **If there is substantial evidence for cardiogenic shock**
  - Pediatric intensivist, cardiologist, ED, and pediatric hospitalist will collaborate to make the best decision about disposition of the patient.
  - If upon evaluation ECMO is recommended, transfer may be required outside of KP system, others may be required to go to LAMC instead of DWN or FON.

Return to Unstable Pathway

Return to Stable Pathway

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APPENDIX 4: LAB INFORMATION

Lab evaluation:

- SARS-COV2 PCR testing may be positive or negative. SARS-COV2 IgG may be detected.
- Sepsis labs (blood culture, lactic acid, DIC panel etc.) and labs to evaluate for incomplete KD:
- CRP, ESR, CBC with differential, BNP, Albumin, ALT, urinalysis, Ferritin, and DIC panel.
- If evidence of LV dysfunction, shock or severe illness, then obtain typical cardiac labs: BNP, Troponin and a full set of electrolytes (to include ionized calcium, magnesium and phosphorous; likely in or requested by PICU).
- The order-sets "COVID-19 MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN (MIS-C) ED SCAL" or "COVID-19 MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN (MIS-C) PEDS IP SCAL Initial Labs section has all these labs.

First Tier labs:

1. CBC with differential
2. Lytes, BUN, creatinine
3. CRP and ESR
4. COVID PCR
5. COVID Serology
6. Blood culture, urinalysis and urine culture

Second Tier labs:

1. Blood culture, urinalysis, urine culture if not already collected
2. EBV serology, ASO, Bacterial GI panel if predominantly GI symptoms
3. Albumin, CK, AST, ALT, T/D bili, PTT, INR
4. Triglycerides, LDH
5. Ferritin
6. BNP and troponin (a high sensitivity troponin is abnormal if > 30 pg/mL)
7. *As of September 2021, D-Dimer and Fibrinogen are not trended or part of 2nd tier unless Rheumatology requests as part of work-up for HLH or other illness.*

Return to Unstable Pathway

Return to Stable Pathway

PEDIATRIC CARE

APPENDIX 5: INPATIENT MANAGEMENT

Suspected MIS-C: ongoing fever, supportive labs showing inflammation, multi-system involvement.
Admit to PICU if shock, hypotension/pressors or concerns for cardiac dysfunction

MANAGEMENT WHILE HOSPITALIZED

Labs:

- Ensure that Tier 2 labs sent: BCx, UA, UCx, EBV, ASO, GI bacterial panel (if predominantly GI sxs), Albumin, CK, AST/ALT, T/D bili, PTT, INR, TG, LDH, Ferritin, BNP, Troponin, and quant gold for TB - please send QUIGs prior to IVIG administration.
- Also ensure COVID-19 PCR and COVID-19 Serology tests have been sent.
- Repeat CBC, manual differential, CRP, lytes, Albumin, Ddimer, Fibrinogen, Ferritin, AST/ALT daily.
- Trend Troponin/BNP daily if abnormal until significantly improved. OR consider q48h until afebrile, if initial set normal (high sensitivity troponin of > 30 pg/mL is abnormal for MIS-C).
- Can space out labs to q48h if considerably downtrending and patient afebrile with clinical improvement.
- (Fibrinogen and D-Dimer are no longer trended as of Sept, 1 2021 unless requested by Rheumatology).

Chest X-ray:

- Recommended at admission.

Echo:

- Recommended at admission.
- Echo timing typically upon admission, within 24 hours, but consider need for echo more urgently if evidence of shock or if clinical concern for myocardial dysfunction in communication with pediatric cardiologist.
- Necessary stabilization and/or treatment of the patient should not be delayed pending a formal echocardiogram.
- Echocardiographic evaluation in MIS-C is typically to evaluate ventricular/valvar function, pericardial effusion, and coronary artery dimensions with indexed Z-scores as needed.
- Consider repeat echocardiograms as inpatient if significant abnormality seen on initial study or worsening clinical status as needed.

EKG:

- 12 lead EKG on admission.
- Consider repeat EKG every 2-3 days per cardiologist/intensivist/hospitalist, especially if abnormality seen on initial EKG.
- Patients with conduction abnormality should be considered for continuous monitoring / telemetry.

Consults:

- Rheumatology
- Infectious Disease
- Cardiology if significant coronary artery involvement, LV dysfunction, arrhythmia, or if clinical myo/pericarditis.
- Neurology and Nephrology in selected cases as needed.

Medications:

- Start IVIG-2 gram/kg. NOTE: IVIG infusion is typically over 12 hours OR LONGER, but may need to be split over 1-2 days depending on consideration of fluid sensitivity and cardiac status. Even if echo is normal, if BNP is elevated then the IVIG should be given over longer period of time. The rate of IVIG infusion should initially be low in MIS-C patients who have shock with myocardial dysfunction, then increased gradually based on tolerance. Monitor for symptoms of fluid overload (respiratory distress, heart failure). Consider continuing low rate and possible diuretic use if significant myocardial dysfunction. Maximum dose of 100 grams.
- Contact Rheumatology for the Solumedrol dosing which is usually given for 3-5 days followed by oral prednisone taper over 3 weeks (TBD when discharge anticipated).
- Antibiotics for at least the first 48 hours until cultures negative

Anticoagulation:

- Start Aspirin 3-5mg/kg daily to max of 81mg daily as long as platelets >80k (to be continued upon discharge)
- Lovenox *may* be recommended by hematology (eg if thrombus or high risk of thrombosis per hematology) or by cardiology (eg if LV EF <35%). If Lovenox recommended, start with goal anti-Xa level of 0.5-1 and continue as outpatient per heme/cardiology recommendations.

Follow-up:

Discharge pending improvement in labs x48hrs, clinical improvement and afebrile x48hrs and tolerating oral steroids

- Will need rheumatology telehealth follow up within 1 week of discharge with repeat labs (CBC, manual differential, CRP, lytes, Albumin, Ddimer, Fibrinogen, Ferritin, AST/ALT, troponin, BNP) ordered to be done outpatient prior to rheum follow up
- Follow-up echocardiograms scheduled at 1-2 weeks and 4-6 weeks after the date of treatment.
- Steroid taper-enough for at least 2-3 weeks
- Aspirin low-dose to be continued until the 4-6 week echocardiogram (or longer if persistent abnormality)
- Follow-up with ID specialist, Rheumatologist, or cardiology, or others as indicated
- *Code patients for now as Hx of COVID19 disease Z86.19 AND Vasculitis, Unspecified I77.6*

Return to Unstable Pathway

Return to Stable Pathway

APPENDIX 6: REFERENCES

- References:
- https://emergency.cdc.gov/han/2020/han00432.asp
- https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.120.048360
- https://www.rcpch.ac.uk/sites/default/files/2020-05/COVID-19-Paediatric-multisystem-%20inflammatory%20syndrome-20200501.pdf
- Muti-System Inflammatory Syndrome in Children (MIS-C) Associated with COVID-19 [formerly Pediatric Multi-System Inflammatory Syndrome (PIMS)] v5.25.20 KP NCAL
- https://www.chop.edu/clinical-pathway/multisystem-inflammatory-syndrome-mis-c-differential-diagnosis
- CDC Clinician Outreach and Communication Activity (COCA) on Clinical Management MIS-C. July 16, 2020. Slides available at: https://emergency.cdc.gov/coca/calls/2020/callinfo_071620.asp
- Clinical Guidance for Pediatric Patients with Multisystem Inflammatory Syndrome in Children (MIS-C) Associated with SARS-CoV-2 and Hyperinflammation in COVID-19, Developed by the ACR MIS-C and COVID-19 Related Hyperinflammation Task Force, Approved June 17, 2020. Document available at: https://www.rheumatology.org/Portals/0/Files/ACR-COVID-19-Clinical-Guidance-Summary-MIS-C-Hyperinflammation.pdf
- Multisystem inflammatory syndrome in children and adults (MIS-C/A): Case definition & guidelines for data collection, analysis, and presentation of immunization safety data. Vogel et al. Vaccine 39 (2021) 3037-3049. Document available at: https://brightoncollaboration.us/wp-content/uploads/2021/07/MIS-CA-vaccine-publication.pdf